PART V

Evidence, Data, and Project Record

The methods, sources, rights, and production record required to inspect and reproduce the Atlas.

CROSSWALK

Nine fields, thresholds, proxies, caveats, and allocations

REGRESSION

Specification, robust uncertainty, comparators, and growth check

PROVENANCE

Exact V-Dem artifact, checksum, and World Bank indicators

PUBLICATION RECORD

Ownership, citation, license, source access, glossary, and colophon

Data Attribution and Source Ownership

Primary Data

• V-Dem (Varieties of Democracy) Country-Year Core, version 15, using 2024 country-year observations. Released component fields: v2cltrnslw, v2juhcind, v2juncind, v2clkill, v2xcl_acjst, v2cltort, v2x_clphy, v2clrspct, and v2x_freexp. Each of the eight V-Dem levers has a distinct operational field or field composite; World Bank indicators supply lever 9.
• World Bank World Development Indicators (WDI) via API — GDP per capita (current US$), population, prevalence of undernourishment, and poverty headcount at $2.15 (2017 PPP). Values are the most recent published as of atlas production (labeled 2026 baseline).
• United Nations Member States list for the 193-country publication scope; repository ISO tables provide operational codes and regional grouping.

Field-level ownership

V-Dem is credited for its institutional measurements. The World Bank is credited for GDP, population, undernourishment, and poverty observations. RTLP scores, RTLDI values, lever classifications, potential-GDP and capital-exclusion estimates, rankings, maps, narratives, and generated data products are attributable to the RTLDI Atlas Project.

Source Framework

The RTLDI equation and the nine RTLP indicators are taken from:
Sid J.A. Hubbard, Causality and Attraction: A Continuum of Steady States (Version 3, May 2026), DOI: 10.5281/zenodo.19468550. The operational crosswalk and binarization thresholds used here are documented in docs/indicator_crosswalk.md of this project.

This PDF was generated programmatically from the open RTLDI ATLAS toolkit. No proprietary or restricted source data were used. V-Dem and World Bank remain the authoritative sources for their observations.

RTLDI Atlas Suite and Public Repository

This PDF atlas was produced by the RTLDI Atlas Suite, the open repository that contains the canonical data crosswalks, Python build pipeline, choropleth map generation, PDF ebook generator, and self-contained HTML atlas generator used for the public release.

Users of this atlas are invited to download the RTLDI Atlas Suite and reproduce the published RTLP scores and RTLDI capital-exclusion projections directly from the public-release V-Dem and World Bank datasets. The suite is designed so NGOs, researchers, policymakers, analysts, and investors can rerun the atlas as new public data releases become available.



RTLDI ATLAS PUBLIC REPOSITORY
https://github.com/SJAH9/rtldi-atlas

The same tools can also be used to generate an RTLDI Atlas from proprietary, internal, or field data. The implementation separates the conceptual nine-lever RTLDI model from the data crosswalk: if an organization has its own trusted country, regional, sectoral, or subnational indicators, it can map those fields into the required nine binary RTLP components and then use the existing suite to compute R, per-capita ΔG, total capital exclusions, lever-level attribution, regional summaries, maps, spreadsheets, PDF reports, and HTML outputs.

For public-data runs, use the documented V-Dem and World Bank workflow. For proprietary-data runs, implement a clear data crosswalk, preserve the nine RTLP component meanings, document all thresholds, and keep the output labels tied to the same canonical RTLDI equation. This keeps forks comparable while allowing local datasets to improve coverage, timeliness, or granularity.

The official public atlas has one canonical pipeline: src.build_atlas produces the master atlas table; src.generate_atlas_ebook.prepare_atlas_data() prepares the shared print/web data; and the PDF/HTML generators consume that same prepared data. Forks should preserve this single-pipeline discipline or clearly identify themselves as experimental derivatives.

Evidence and Transparency Roadmap

The pages that follow mirror the web atlas's methods and source views. Each lever page states the RTLP question, source field, threshold, operational proxy, caveat, and global bounded allocation. Regression pages report specification, robust uncertainty, statistical significance, comparators, and a non-significant one-year growth check. Dataset pages identify the exact V-Dem artifact and World Bank indicators used for reproduction.

This evidence should be read with the interpretation limits in Part I. Lever-level allocations partition a bounded national total; they are not independent causal estimates. Cross-sectional significance does not establish a reform counterfactual.

The Nine Levers: Executable Crosswalk

This section mirrors the web atlas lever pages. It reports the global annual capital exclusions attributed to each lever and the exact fields and thresholds used by the executable 2026 pipeline. The lever amounts allocate each country's bounded total across its missing components; they are not nine independent causal estimates and should not be summed without the shared cap.

LEVER 1 OF 9

Legal Protections

$3.16T global annual exclusions

Present when ordinary law is applied consistently to state agents as well as everyone else. Absent when power can act without predictable legal constraint. Multiplies GDP by lowering the risk of arbitrary loss; people and businesses invest and trade more when they know the rules will be enforced fairly. Annual cost is low.

RTLP QUESTION

Does the entity have provisions explicitly protecting the right to life?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2cltrnslw
Binary rule: v2cltrnslw >= 2.0
Operational proxy: Transparent laws with predictable enforcement

Transparency note: Operational de-facto proxy; it is not a direct constitutional right-to-life field.

LEVER 2 OF 9

Independent Judiciary

$3.22T global annual exclusions

Present when there is a judiciary capable of upholding right-to-life laws even against the state. These protections represent opportunities for property rights and contracts to support reliable capital flows to productive uses. Annual cost is low relative to the scale unlocked.

RTLP QUESTION

Is there an independent judiciary capable of upholding right-to-life laws?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2juhcind, v2juncind
Binary rule: mean(v2juhcind, v2juncind) >= 2.0
Operational proxy: High-court and lower-court independence

Transparency note: The two available court estimates are averaged when either is present.

LEVER 3 OF 9

Law Enforcement Accountability

$1.45T global annual exclusions

Present when law enforcement agencies are accountable for unlawful killings and violence. These protections represent opportunities to support greater economic scale by reducing arbitrary risk that can deter investment, labor mobility, and everyday economic activity. Annual cost is primarily institutional oversight.

RTLP QUESTION

Are law-enforcement agencies accountable for unlawful killings?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2clkill
Binary rule: v2clkill >= 2.0
Operational proxy: Freedom from political killings

Transparency note: Outcome/accountability proxy for unlawful state killings.

LEVER 4 OF 9

Protection Against Arbitrary Detention

$1.01T global annual exclusions

Present when arbitrary detention is prohibited with real legal recourse. These protections represent opportunities for individuals and families to plan, work, and invest with greater security, supporting scale in the economy. Annual cost is low.

RTLP QUESTION

Is arbitrary detention prohibited with legal recourse?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2xcl_acjst
Binary rule: v2xcl_acjst >= 0.5
Operational proxy: Access to justice index

Transparency note: There is no direct arbitrary-detention field in the current Core crosswalk; access to justice is used as the closest recourse proxy.

LEVER 5 OF 9

Freedom from Torture and Inhumane Treatment

$3.31T global annual exclusions

Present when the state has effective measures to prevent torture. These protections represent opportunities to support human capital and cooperation. Annual cost is institutional and training: an investment toward realizing greater scale.

RTLP QUESTION

Are measures in place to prevent torture and inhumane treatment?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2cltort
Binary rule: v2cltort >= 2.0
Operational proxy: Freedom from torture

Transparency note: Direct substantive proxy, converted from the model estimate to a binary atlas score.

LEVER 6 OF 9

Civilian Protection in Conflict Zones

$834.8B global annual exclusions

Present when mechanisms exist to protect civilians during conflict. These protections represent opportunities to preserve the productive capacity and future of entire regions. Annual cost is primarily military discipline and rules of engagement.

RTLP QUESTION

In conflict areas, are mechanisms protecting civilians?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2x_clphy
Binary rule: v2x_clphy >= 0.5
Operational proxy: Physical violence index

Transparency note: High-level physical-integrity proxy, distinct from lever 3's political-killings component.

LEVER 7 OF 9

Access to Justice

$3.39T global annual exclusions

Present when ordinary people can bring serious claims and have them heard fairly. These conditions represent opportunities for contracts and rights to function for a wider share of the population. Annual cost is court capacity and legal aid.

RTLP QUESTION

Can individuals obtain fair remedies for right-to-life violations?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2clrspct
Binary rule: v2clrspct >= 2.0
Operational proxy: Rigorous and impartial public administration

Transparency note: Administrative impartiality is the distinct operational proxy for whether remedies are applied fairly.

LEVER 8 OF 9

Freedom of Expression and Whistleblower Protections

$993.5B global annual exclusions

Present when laws protect people who report corruption from retaliation. These conditions represent opportunities for information to flow more accurately and for public resources to be used more effectively. Annual cost is low: mainly passing and enforcing the law.

RTLP QUESTION

Are expression and whistleblower rights upheld?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2x_freexp
Binary rule: v2x_freexp >= 0.5
Operational proxy: Freedom of expression and alternative information index

Transparency note: Expression is the operational proxy; the current crosswalk has no direct whistleblower-law variable.

LEVER 9 OF 9

Socioeconomic Conditions

$1.91T global annual exclusions

Present when a society makes the deliberate choice to maintain a minimum floor against extreme deprivation. This reflects political will rather than just economic outcomes. When present, it increases the share of the population able to participate productively, expanding overall economic scale. Annual cost is a political decision about minimum security standards.

RTLP QUESTION

Are food and minimum material conditions adequate to sustain life?

FIELDS AND SCORING RULE

Source: World Bank WDI
Field(s): SN.ITK.DEFC.ZS, SI.POV.DDAY
Binary rule: undernourishment <= 5% AND poverty headcount <= 10%
Operational proxy: Prevalence of undernourishment and international-poverty headcount

Transparency note: Missing values are conservatively treated as failure. This is an outcome-based basic-needs proxy, not a V-Dem field.

GDP Regression, Significance, and Eta

The canonical atlas model is a population-weighted cross-sectional regression of log GDP per capita on activated-lever count. It estimates association with the level of GDP per capita, not a causal time-series growth effect.

ETA	95% ROBUST INTERVAL	MODEL FIT
0.356	5.7% to 73.9%	R2 0.242
exp(beta) - 1	n = 187	p = 0.0177

Specification: log(GDP per capita) = intercept + beta x activated-lever count + error. The fitted log slope is 0.304 with robust standard error 0.127. Eta is exp(beta)-1 = 0.356; the operational atlas rounds this to 0.36 and applies Delta G = min(eta x (1-R) x G0, 0.25 x G0).

INTERPRETATION LIMIT

Statistical significance is not causal identification. Omitted variables, reverse causation, proxy validity, binary thresholds, and source-year mismatch remain. The 25% cap limits attribution but does not resolve those identification problems.

Regression Comparators and Growth Check

Single predictor	n	Slope	R2	p
Population	187	-0.161	0.063	0.000188
Undernourishment	159	-0.094	0.468	1.19e-14
Extreme-poverty headcount	61	-0.115	0.351	2.49e-05

Available growth falsification check: 2023-2024 current-US$ GDP-per-capita change on lever count, unweighted OLS. n=179; slope=-0.193 percentage points per lever; R2=0.004; p=0.165. This one-year check is not statistically significant. The atlas therefore describes a cross-sectional GDP-level association and does not claim that the nine levers predict one-year GDP growth.

Comparator models are descriptive single-predictor checks. Undernourishment and poverty are development outcomes and inputs to lever 9, not independent causal controls; missing data change their sample sizes.

V-Dem Dataset and Reproduction

The corrected 2026 score table uses V-Dem Country-Year Core v15 and 2024 country-year observations. The current update workflow targets Country-Year Core v16. The repository does not redistribute the large raw source artifact.

SHA-256 for the exact v15 artifact: 4b715190a1e351a81daf11cbe6f9f77e67d8914f0ed612b93e2d8fba5523e540

For exact replication, download the v15 artifact identified above and export the required columns as data/raw/V-Dem-CY-Core-v15.csv. For a current-data update, download Core v16, then run python3 -m src.build_atlas --year 2026 --eta 0.36.

OFFICIAL V-DEM V15 ARCHIVE
https://www.v-dem.net/data/dataset-archive/

EXACT V-DEM V15 ARTIFACT
https://raw.githubusercontent.com/vdeminstitute/vdemdata/V15/data/vdem.RData

CURRENT V-DEM DATASET
https://www.v-dem.net/data/the-v-dem-dataset/

V-DEM V16 CODEBOOK
https://www.v-dem.net/documents/70/codebook_v16.pdf

V-Dem's unsuffixed expert-coded fields are measurement-model point estimates on a standardized interval scale, not the original ordinal coder categories. The atlas cutoffs are project operationalizations. A v16 rebuild may revise historical scores and is therefore an update rather than guaranteed bit-for-bit replication. The URL and checksum above identify the exact official v15 artifact used by this build.

World Bank Dataset and Reproduction

World Development Indicators supply NY.GDP.PCAP.CD (GDP per capita, current US$), SP.POP.TOTL (population), SN.ITK.DEFC.ZS (undernourishment), and SI.POV.DDAY (international-poverty headcount). GDP is G0 and the regression outcome; population is the regression weight and aggregation factor; the last two fields populate lever 9.

The pipeline fetches the public API through wbgapi and stores processed snapshots under data/processed/.



WORLD BANK DATA PORTAL
https://data.worldbank.org/



WORLD BANK PUBLIC API
https://api.worldbank.org/v2/

The 2026 baseline is the freshest published value available at build time, not a claim that every value was observed in 2026. Missing undernourishment or poverty is conservatively scored as lever 9 absent. Current-US$ GDP supports current-dollar totals but also reflects price-level and exchange-rate movement.

Citation, Licensing, and Reuse

CITE ATLAS OUTPUTS

RTLDI Atlas Project. (2026). RTLDI Atlas of UN Member States [Data set, software, and atlas]. Sid J.A. Hubbard, Principal Investigator.

ATLAS REPOSITORY
https://github.com/SJAH9/rtldi-atlas

CITE THE SOURCE THEORY

Hubbard, Sid J.A. (2026). Causality and Attraction: A Continuum of Steady States, Volume I, Version 3.

SOURCE THEORY DOI
https://doi.org/10.5281/zenodo.19468550

CITE SOURCE OBSERVATIONS

For institutional observations, cite the V-Dem Institute and the v15 dataset/codebook appropriate to the observation. For GDP, population, undernourishment, and poverty observations, cite the relevant World Development Indicators series and access date. Do not attribute RTLDI Project transformations or projections to V-Dem or the World Bank.

LICENSE BOUNDARY

Atlas repository materials are available under the MIT License. Preserve the copyright and license notice when redistributing licensed materials. The source books and theoretical framework retain their separately stated rights.

MIT LICENSE
https://github.com/SJAH9/rtldi-atlas/blob/main/LICENSE

PART V / REFERENCE

Glossary and Source Access

Definitions for the Atlas, with direct access to the executable crosswalk, source code, canonical records, and reproduction commands.

https://github.com/SJAH9/rtldi-atlas

The glossary follows this title page. Scan to inspect the code and project-generated data; V-Dem and World Bank remain authoritative for their observations.

Scan with a phone camera or use the printed URL. QR destination verified at production time.

Glossary of Terms

2026-fresh G0:
: The most recent published GDP per capita values used as the dynamic baseline for the 2026 edition (even when RTLP components are from 2024).

Access to Justice:
: Ordinary people can bring serious claims and have them heard fairly.

AI-assisted production:
: The disclosed use of AI and data-science tools in research, software development, analysis, validation, and publication under the direction and responsibility of the Principal Investigator.

Arbitrary Detention:
: Core protection against state overreach (RTLP indicator #4); current implementation level contributes to associated capital exclusions in regions.

Attributable capital exclusions:
: The allocation of a country or region's bounded total across missing levers, using $\eta/9 \times G_0 \times$ population and each country's cap ratio. Current operational η=0.36. These allocations are not independent causal estimates.

Binarization / threshold:
: The documented conversion of continuous V-Dem and World Bank values into the binary 0/1 scores for the nine indicators (e.g., ≥2.0 on 0-4 scales, ≤5 % undernourishment + ≤10 % poverty for #9).

Capital exclusion:
: The Atlas term for bounded lost potential GDP associated with missing RTLP protections under the stated cross-sectional model. It is an estimate, not observed money or a guaranteed reform dividend.

Causal projection:
: A projection of how changing an enclosure alters the set of reachable outcomes. It identifies an enabled or disabled pathway without guaranteeing the timing, magnitude, or realization of a particular response.

Causality and Attraction:
: The source monograph (Hubbard 2026V3, Zenodo 19468550) that supplies the RTLDI equation, the nine RTLP indicators, the nested-enclosures paradigm, and the cartographic critique of distortion. This atlas extends the source RTLDI to quantify capital exclusions.

Choropleth:
: Thematic map in which countries or regions are shaded by enclosure strength R (0–1) using the RTLDI atlas scale; primary visual for the global, regional, and focus views.

Civilian Protection in Conflict Zones:
: Mechanisms protect civilians during conflict.

Contextual cap:
: The rule limiting projected capital exclusions to 25% of G_0 so the nine protections are not assigned more than one quarter of observed GDP. The cap limits attribution but does not establish causality.

Country view:
: The country-only choropleth on each nation page, fitted to the national boundary and shaded by enclosure strength R without using the regional context map.

Cross-sectional association:
: A relationship measured across countries at one analytical period. It can be statistically significant without showing that changing one variable will cause the other to change.

Crosswalk:
: The explicit mapping from the nine verbatim RTLP questions to observable V-Dem variables and World Bank statistics, with defined binarization thresholds.

ΔG (delta G):
: Annual per-capita capital exclusions (lost potential GDP) associated with the current level of the nine levers. Core output of the RTLDI equation.

DOI:
: Digital Object Identifier: a persistent identifier for a publication. The Atlas source framework is identified by doi:10.5281/zenodo.19468550.

Economic Intelligence (tool):
: The use of R scores, component breakdowns, and attributable capital exclusions figures as practical information for governments, businesses, and analysts to identify high-leverage institutional opportunities.

Enclosure (nested causal / human):
: The source document's central metaphor: the interdependent, layered structures of protection whose strength or weakness shapes higher-order economic and social steady states.

η (eta):
: The sensitivity coefficient (current data-driven default 0.36 from the corrected population-weighted 2026 UN cross-section). In the bounded model $\Delta G = \min(\eta(1 - R) \times G_0, 0.25 \times G_0)$; the cross-sectional premium per indicator, then capped so the nine RTLP indicators are never credited with more than 25% of observed G_0 after other determinants are given due weight. (Source document used a more conservative 0.05 structural value.)

Existence of Legal Protections:
RTLP indicator #1; the foundational de-jure and de-facto right to life.

Feasible set:
The outcomes and actions that remain institutionally possible under the current enclosures. Activating a lever can add pathways to this set without forcing actors to use them immediately.

Freedom from Torture and Inhumane Treatment:
Measures are in place to prevent torture and inhumane treatment.

Freedom of Expression and Whistleblower Protections:
Expression and whistleblower rights are protected from retaliation.

Full Data View:
The web table exposing one record per UN Member State with RTLP, lever, GDP, population, potential-GDP, and capital-exclusion fields.

G_0 (G-zero):
Baseline GDP per capita (World Bank, current US$). The 'current' economic size against which capital exclusions are measured.

Independent Judiciary:
Courts can uphold right-to-life laws, including against the state.

Institutional option value:
The value created when an effective protection makes a beneficial action or remedy available. Availability can be established by the protection even when the eventual uptake and financial return remain uncertain.

Law Enforcement Accountability:
Law enforcement agencies are accountable for unlawful killings and violence.

Legal Protections:
Ordinary law applies consistently to state agents and everyone else.

Linked open data:
Atlas data published in RDF/XML with stable concepts and relationships so machines can consume and connect the observations semantically.

Member-nations table:
The compact table on each regional summary page that lists every country in the region together with its R, G0, population, and capital exclusions, plus a bottom row of regional totals.

MIT License:
The permissive open-source license governing Atlas repository code, pipeline, documentation, and generated outputs. It does not relicense the source books or theoretical framework.

Mollweide:
Equal-area pseudocylindrical map projection used for global and regional choropleths. Nation profiles use fitted country views for legibility.

Operational proxy:
An available measured field used to represent a conceptual lever when no source variable directly measures the full RTLP question.

Outer enclosure:
A legal, institutional, or social boundary that protects and constrains actions inside a modeled system. In the whistleblower example, credible protection surrounds the reporting pathway and changes what insiders can safely do.

p-value:
The probability, under the fitted null model, of observing a test statistic at least as extreme as the one reported. It is not the probability that the model is true or causal.

Population weighting:
Regression or aggregation weights proportional to national population, so the fitted relationship or mean reflects people rather than treating every country as equally sized.

Protection Against Arbitrary Detention:
Arbitrary detention is prohibited with real legal recourse.

Provenance:
The documented origin, version, transformation path, and checksum needed to understand or reproduce a data product.

QR gateway:
A full-page print-to-digital bridge containing a high-redundancy QR symbol and its visible destination URL. The URL remains usable without QR scanning.

R (RTLP score):
The arithmetic mean (0–1) of the nine binary Right-to-Life Protection lever values. R=1 means all nine operational thresholds are met.

R-squared (R2):
The share of outcome variation described by a fitted regression in its sample. It is a model-fit statistic, not evidence of causality.

Right to Life:
The overarching protected interest; the nine indicators operationalize equal and effective protection against arbitrary interruption.

Robust standard error:
An uncertainty estimate designed to remain valid under heteroskedastic regression residuals; used in the Atlas regression interval and significance test.

RTLDI:
: Right to Life Deficit Index — the source framework operationalized here to report RTLP scores and bounded capital-exclusion estimates.

RTLDI atlas R scale:
: The shared perceptual color scale used for every enclosure-strength choropleth in the atlas: deep low-R tones through teal midrange values to warm high-R values.

RTLP:
: Right-to-Life Protection. In the Atlas, RTLP refers to the nine-component protection score R and its component classifications.

Socioeconomic Conditions:
: A minimum floor against extreme deprivation is maintained.

Source ownership:
: The attribution boundary separating V-Dem and World Bank observations from RTLDI Atlas Project classifications, transformations, projections, maps, and narratives.

Steady state:
: A balanced biological, economic, or social condition that the framework argues is supported or undermined by the strength of the nine life protections and the resulting causal enclosures.

The Nine Levers:
: The nine conceptual protections averaged to produce R. Each is populated by a distinct operational field or field composite in the corrected pipeline.

Torture and Inhumane Treatment:
: RTLP indicator #5; one of the larger contributors to global capital exclusions, highlighting physical integrity protections.

Trend (up to 3 years):
: A text summary of available GDP-driven projection points with R held fixed at its latest V-Dem value. It is not a measured RTLP time series.

UN region / regional summary:
: One of the 22 geographic groupings used for aggregates, focused choropleths, member-nations tables (with R / G0 / population / capital exclusions + REGIONAL TOTAL row), and 9-lever breakdowns.

V-Dem:
: Varieties of Democracy project. Supplies the expert-coded governance and civil liberties indicators for eight of the nine RTLP components.

Weighted seven-signal equivalence:
: The algebraic rewriting of the former duplicated-proxy nine-row score as seven unique signals while retaining two-ninths weights for the two repeated pairs. It is not an unweighted seven-variable model.

Credits and Acknowledgments

Principal Investigator and Production
Sid J.A. Hubbard conceived, wrote, directed, and engineered the RTLDI Atlas Project and is responsible for its methods, crosswalk decisions, interpretation, software direction, and publication. The project was produced with assistance from multiple AI and data-science tools under the production disclosure stated in Part I.

Conceptual Framework
Causality and Attraction: A Continuum of Steady States, Volume I, Version 3 — Sid J.A. Hubbard.

Institutional Data
V-Dem Institute — Country-Year Core v15, 2024 observations. SHA-256 4b715190a1e351a81daf11cbe6f9f77e67d8914f0ed612b93e2d8fba5523e540.

Economic and Population Data
World Bank World Development Indicators through the public API.

Country Scope
United Nations Member States.

CONCEPTUAL FRAMEWORK DOI
https://doi.org/10.5281/zenodo.19468550

EXACT V-DEM V15 ARTIFACT
https://raw.githubusercontent.com/vdeminstitute/vdemdata/V15/data/vdem.RData

V-DEM INSTITUTE
https://www.v-dem.net/

WORLD BANK DATA
https://data.worldbank.org/

WORLD BANK API
https://api.worldbank.org/v2/

UNITED NATIONS MEMBER STATES
https://www.un.org/en/about-us/member-states

OPEN-SOURCE PRODUCTION STACK
Python, pandas, NumPy, SciPy, Plotly, Kaleido, fpdf2, PyPDF2, Pillow, qrcode, openpyxl, and wbgapi. Plotly's geographic rendering uses public map geometry derived from established open cartographic sources, including Natural Earth. Each project retains its own license and attribution requirements.

Colophon and Print Specification

Format

A4 portrait (210 × 297 mm), designed without required bleed. All essential text, QR quiet zones, maps, folios, and rules remain inside the trim safety area. Section gateways and covers use white trim edges so the file can be printed without edge-to-edge finishing.

Typography and graphics

Unicode-capable TrueType fonts are embedded by the PDF generator. Global and regional maps are rendered from the canonical atlas data; QR symbols are generated at high resolution with error-correction level H. Rules and text remain vector content.

Assembly

The publication is generated as modular front, regional, nation, Eurozone, and evidence PDFs, then assembled in canonical order. Final processing adds continuous physical folios, document metadata, a two-page reading layout preference, and PDF outline navigation.

Production note

This general-purpose print file uses a device-independent digital workflow suitable for office, digital, and print-on-demand production. A commercial offset printer may apply its required CMYK output intent, paper profile, imposition, and trapping during prepress.

REPOSITORY
https://github.com/SJAH9/rtldi-atlas

WEB ATLAS
https://sjah9.github.io/rtldi-atlas/outputs/html-atlas/

INTERACTIVE MAP
https://sjah9.github.io/rtldi-atlas/outputs/html-atlas/global_interactive.html

FULL DATA VIEW
https://sjah9.github.io/rtldi-atlas/outputs/html-atlas/full_data.html